CS 395T Homework 3

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Problem 1

(i)

Proof. The feasible set $\mathcal{F}_i := \{\mathbf{x} \in \mathbb{R}^n \mid \mathbf{A}^\top \mathbf{x} = \mathbf{a}_i\}$ is nonempty: since $\mathbf{a}_i$ is the i-th row of $\mathbf{A}$, we have $\mathbf{A}^\top \mathbf{e}_i = \mathbf{a}_i$, so $\mathbf{e}_i \in \mathcal{F}_i$.

Decompose $\mathbb{R}^n = \mathrm{col}(\mathbf{A}) \oplus \ker(\mathbf{A}^\top)$ orthogonally (by the fundamental theorem of linear algebra). Any $\mathbf{x} \in \mathcal{F}_i$ splits as $\mathbf{x} = \mathbf{x}_R + \mathbf{x}_N$ with $\mathbf{x}_R \in \mathrm{col}(\mathbf{A})$ and $\mathbf{x}_N \in \ker(\mathbf{A}^\top)$. Then

$$\mathbf{A}^\top \mathbf{x} = \mathbf{A}^\top \mathbf{x}_R + \mathbf{A}^\top \mathbf{x}_N = \mathbf{A}^\top \mathbf{x}_R = \mathbf{a}_i,$$

so $\mathbf{x}_R \in \mathrm{col}(\mathbf{A}) \cap \mathcal{F}_i$ as well. By orthogonality,

$$\|\mathbf{x}\|_2^2 = \|\mathbf{x}_R\|_2^2 + \|\mathbf{x}_N\|_2^2 \geq \|\mathbf{x}_R\|_2^2,$$

with equality iff $\mathbf{x}_N = \mathbf{0}$. Thus $\min_{\mathbf{x} \in \mathcal{F}_i} \|\mathbf{x}\|_2^2 = \min_{\mathbf{x} \in \mathrm{col}(\mathbf{A}) \cap \mathcal{F}_i} \|\mathbf{x}\|_2^2$.

There is a unique $\mathbf{x}^* \in \mathrm{col}(\mathbf{A}) \cap \mathcal{F}_i$: write $\mathbf{x}^* = \mathbf{A}\mathbf{y}$ for $\mathbf{y} \in \mathbb{R}^d$; the constraint $\mathbf{A}^\top \mathbf{x}^* = \mathbf{a}_i$ gives $\mathbf{A}^\top \mathbf{A}\mathbf{y} = \mathbf{a}_i$. Since $\mathbf{a}_i \in \mathrm{col}(\mathbf{A}^\top) = \mathrm{col}(\mathbf{A}^\top \mathbf{A})$, the minimum-norm solution is $\mathbf{y} = (\mathbf{A}^\top \mathbf{A})^\dagger \mathbf{a}_i$, yielding

$$\mathbf{x}^* = \mathbf{A}(\mathbf{A}^\top \mathbf{A})^\dagger \mathbf{a}_i.$$

The minimum value is

$$\begin{aligned} \min_{\mathbf{x} \in \mathcal{F}_i} \|\mathbf{x}\|_2^2 = \|\mathbf{x}^*\|_2^2 &= \mathbf{a}_i^\top (\mathbf{A}^\top \mathbf{A})^\dagger \mathbf{A}^\top \mathbf{A} (\mathbf{A}^\top \mathbf{A})^\dagger \mathbf{a}_i \\ &= \mathbf{a}_i^\top (\mathbf{A}^\top \mathbf{A})^\dagger \mathbf{a}_i = \boldsymbol{\tau}_i(\mathbf{A}), \end{aligned}$$

where the second equality uses the pseudoinverse identity $\mathbf{M}^\dagger \mathbf{M} \mathbf{M}^\dagger = \mathbf{M}^\dagger$. $\square$

(ii)

Proof. Let $\mathbf{A} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$ be the thin SVD of $\mathbf{A}$, where $r := \mathrm{rank}(\mathbf{A})$, $\mathbf{U} \in \mathbb{R}^{n \times r}$ and $\mathbf{V} \in \mathbb{R}^{d \times r}$ have orthonormal columns, and $\boldsymbol{\Sigma} = \mathrm{diag}(\sigma_1, \ldots, \sigma_r)$ with $\sigma_j > 0$. Define

$$\mathbf{P} := \mathbf{U}\mathbf{U}^\top \in \mathbb{R}^{n \times n}.$$

$\mathbf{P}$ is an orthogonal projection matrix:

$$\mathbf{P}^2 = \mathbf{U}\mathbf{U}^\top\mathbf{U}\mathbf{U}^\top = \mathbf{U}\mathbf{I}_r\mathbf{U}^\top = \mathbf{P},$$

and $\mathbf{P}^\top = \mathbf{P}$. Moreover, $\mathbf{v}^\top\mathbf{P}\mathbf{v} = \|\mathbf{U}^\top\mathbf{v}\|_2^2 \geq 0$ for all $\mathbf{v}$, so $\mathbf{P} \in \mathbb{S}^{n\times n}_{\succeq 0}$.

$\mathbf{P}_{ii} = \boldsymbol{\tau}_i(\mathbf{A})$ for all $i \in [n]$: The pseudoinverse is $(\mathbf{A}^\top\mathbf{A})^\dagger = \mathbf{V}\boldsymbol{\Sigma}^{-2}\mathbf{V}^\top$. Using $\mathbf{a}_i = \mathbf{A}^\top\mathbf{e}_i = \mathbf{V}\boldsymbol{\Sigma}\mathbf{U}^\top\mathbf{e}_i$,

$$\begin{aligned}\boldsymbol{\tau}_i(\mathbf{A}) = \mathbf{a}_i^\top(\mathbf{A}^\top\mathbf{A})^\dagger\mathbf{a}_i &= \mathbf{e}_i^\top\mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top \cdot \mathbf{V}\boldsymbol{\Sigma}^{-2}\mathbf{V}^\top \cdot \mathbf{V}\boldsymbol{\Sigma}\mathbf{U}^\top\mathbf{e}_i \\ &= \mathbf{e}_i^\top\mathbf{U}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^{-2}\boldsymbol{\Sigma}\mathbf{U}^\top\mathbf{e}_i = \mathbf{e}_i^\top\mathbf{U}\mathbf{U}^\top\mathbf{e}_i = \mathbf{P}_{ii},\end{aligned}$$

where we used $\mathbf{V}^\top\mathbf{V} = \mathbf{I}_r$ twice. □

Problem 2

Proof. By the spectral theorem and Lemma 6 (Part VI), the matrix inequality $(1-\epsilon)\mathbf{M} \preceq p(\mathbf{M})^2 \preceq (1+\epsilon)\mathbf{M}$ holds iff

$$(1-\epsilon)\lambda \leq p(\lambda)^2 \leq (1+\epsilon)\lambda \quad \text{for all } \lambda \in [\mu, L].$$

It therefore suffices to construct a polynomial p of degree $O(\sqrt{\kappa}\log\frac{\kappa}{\epsilon})$ satisfying $|p(\lambda)^2 - \lambda| \leq \epsilon\lambda$ for all $\lambda \in [\mu, L]$.

Chebyshev approximation of $\sqrt{\lambda}$. Map $[\mu, L]$ to $[-1, 1]$ via $\lambda(w) = \frac{L+\mu}{2} + \frac{L-\mu}{2}w$, and define $f(w) := \sqrt{\lambda(w)}$, which is real-valued on $[-1, 1]$. The branch point $\lambda = 0$ corresponds to $w^\star = -\frac{L+\mu}{L-\mu} = -\frac{\kappa+1}{\kappa-1} < -1$.

Set $\rho := 1 + \frac{1}{\sqrt{\kappa}}$. The leftmost real point of the Bernstein ellipse E_ρ is at $w = -\frac{\rho+\rho^{-1}}{2}$. Using $\frac{\rho+\rho^{-1}}{2} = 1 + \frac{(\rho-1)^2}{2\rho} = 1 + \frac{1}{2\kappa+2\sqrt{\kappa}}$, we verify

$$\frac{\rho+\rho^{-1}}{2} = 1 + \frac{1}{2\kappa+2\sqrt{\kappa}} < 1 + \frac{2}{\kappa-1} = \frac{\kappa+1}{\kappa-1} = |w^\star|,$$

where the inequality holds since $\kappa - 1 < 4\kappa + 4\sqrt{\kappa}$ for all $\kappa > 1$. Hence $w^\star$ lies outside E_ρ.

The affine map $\lambda(w)$ sends the interior of E_ρ (simply connected) to a simply connected region $D \subset \mathbb{C}$. Since the leftmost real point of D is

$$\frac{L+\mu}{2} - \frac{L-\mu}{2} \cdot \frac{\rho+\rho^{-1}}{2} = \mu - \frac{(L-\mu)}{4\kappa+4\sqrt{\kappa}} > 0,$$

we have $0 \notin D$. By the given analytic continuation, $\sqrt{\lambda}$ extends analytically to D, so f is analytic on the interior of E_ρ.

Bounding $|f|$ on E_ρ. For $w \in E_\rho$, $|w| \leq \frac{\rho+\rho^{-1}}{2}$, so

$$|\lambda(w)| \leq \frac{L+\mu}{2} + \frac{L-\mu}{2} \cdot \frac{\rho+\rho^{-1}}{2} = L + \frac{L-\mu}{4\kappa+4\sqrt{\kappa}} \leq 2L.$$

Therefore $M := \sup_{w\in E_\rho} |f(w)| \leq \sqrt{2L}$.

Truncation error. By Theorem 3 (Part VII), the Chebyshev coefficients of f satisfy $|a_i| \le 2M\rho^{-i}$ for $i \ge 1$. By Lemma 3 (Part VII), the degree-Δ Chebyshev truncation $q_\Delta(w) := \sum_{i=0}^{\Delta} a_i T_i(w)$ satisfies

$$\sup_{w\in[-1,1]} |f(w) - q_\Delta(w)| \le \sum_{i>\Delta} |a_i| \le \frac{2M\rho^{-\Delta}}{\rho - 1}.$$

Define $p(\lambda) := q_\Delta\left(\frac{2\lambda-(L+\mu)}{L-\mu}\right)$, a degree-$\Delta$ polynomial in λ, so that

$$E := \sup_{\lambda\in[\mu,L]} |p(\lambda) - \sqrt{\lambda}| \le \frac{2\sqrt{2L}\cdot\rho^{-\Delta}}{1/\sqrt{\kappa}} = 2\sqrt{2}\,\frac{L}{\sqrt{\mu}}\,\rho^{-\Delta}.$$

Choosing the degree. Set $\Delta := \lceil 2\sqrt{\kappa}\log(6\sqrt{2}\,\kappa/\epsilon)\rceil$. Since $\log\rho = \log(1+1/\sqrt{\kappa}) \ge \frac{1}{2\sqrt{\kappa}}$ (using $\log(1+x) \ge x/2$ for $x \in (0,1]$),

$$\rho^{-\Delta} \le \exp\left(-\frac{\Delta}{2\sqrt{\kappa}}\right) \le \exp\left(-\log\frac{6\sqrt{2}\,\kappa}{\epsilon}\right) = \frac{\epsilon}{6\sqrt{2}\,\kappa}.$$

Substituting:

$$E \le 2\sqrt{2}\,\frac{L}{\sqrt{\mu}}\cdot\frac{\epsilon}{6\sqrt{2}\,\kappa} = \frac{\epsilon L}{3\kappa\sqrt{\mu}} = \frac{\epsilon\sqrt{\mu}}{3},$$

where the last step uses $L/\kappa = \mu$.

Converting to the quadratic bound. For any $\lambda \in [\mu, L]$, since $E \le \frac{\epsilon\sqrt{\mu}}{3} \le \frac{\sqrt{\mu}}{3} \le \sqrt{\lambda}$:

$$|p(\lambda) + \sqrt{\lambda}| \le E + 2\sqrt{\lambda} \le 3\sqrt{\lambda}.$$

Therefore

$$|p(\lambda)^2 - \lambda| = |p(\lambda) - \sqrt{\lambda}|\,|p(\lambda) + \sqrt{\lambda}| \le \frac{\epsilon\sqrt{\mu}}{3}\cdot 3\sqrt{\lambda} = \epsilon\sqrt{\mu\lambda} \le \epsilon\lambda,$$

where the last inequality uses $\sqrt{\mu} \le \sqrt{\lambda}$. This gives $(1-\epsilon)\lambda \le p(\lambda)^2 \le (1+\epsilon)\lambda$ for all $\lambda \in [\mu, L]$, and $\Delta = O(\sqrt{\kappa}\log\frac{\kappa}{\epsilon})$. □

Problem 3

If $\|\mathbf{v}\|_1 \le R$, then $\mathbf{v}$ itself is feasible and the objective is zero, so the optimum is $\mathbf{x}^* = \mathbf{v}$. Assume henceforth that $\|\mathbf{v}\|_1 > R$.

Since $\mathbf{v} \in \mathbb{R}^d_{\ge 0}$, the optimal $\mathbf{x}^*$ satisfies $\mathbf{x}^* \ge \mathbf{0}$: replacing any negative component x_i^* by 0 decreases $\|x_i - v_i\|$ (since $v_i \ge 0$) and decreases $\|\mathbf{x}\|_1$, so it strictly improves the objective while remaining feasible. With $\mathbf{x} \ge \mathbf{0}$, the constraint $\|\mathbf{x}\|_1 \le R$ becomes $\sum_i x_i \le R$, so the problem reduces to

$$\min_{\mathbf{x}\ge\mathbf{0},\,\sum_i x_i \le R} \frac{1}{2}\|\mathbf{x} - \mathbf{v}\|_2^2.$$

Claim: The solution is $x_i^* = (v_i - \lambda)_+$ for the unique $\lambda \ge 0$ satisfying $\sum_{i=1}^d (v_i - \lambda)_+ = R$.

Proof. The Lagrangian is

$$L(\mathbf{x}, \lambda, \boldsymbol{\mu}) = \frac{1}{2}\|\mathbf{x} - \mathbf{v}\|_2^2 + \lambda\Big(\sum_i x_i - R\Big) - \sum_i \mu_i x_i,$$

with multipliers $\lambda \geq 0$ and $\mu_i \geq 0$. The KKT stationarity condition in coordinate i is

$$x_i - v_i + \lambda - \mu_i = 0, \qquad \text{i.e.,} \quad x_i = v_i - \lambda + \mu_i.$$

Complementary slackness requires $\mu_i x_i = 0$ for each i:

Case 1: $x_i > 0$. Then $\mu_i = 0$, so $x_i = v_i - \lambda$. Since $x_i > 0$, this requires $v_i > \lambda$.

Case 2: $x_i = 0$. From stationarity, $0 = v_i - \lambda + \mu_i$, so $\mu_i = \lambda - v_i$. The requirement $\mu_i \geq 0$ gives $v_i \leq \lambda$.

Combining both cases: $x_i^* = \max(v_i - \lambda, 0) = (v_i - \lambda)_+$.

Since $\|\mathbf{v}\|_1 > R$, the constraint $\sum_i x_i \leq R$ is active at the optimum (if it were inactive, then $\lambda = 0$ by complementary slackness, giving $x_i = v_i$ and $\sum_i x_i = \|\mathbf{v}\|_1 > R$, a contradiction). So $\lambda > 0$ and

$$\sum_{i=1}^{d} (v_i - \lambda)_+ = R.$$

The left side is continuous and strictly decreasing in λ on $[0, \max_i v_i]$, equals $\|\mathbf{v}\|_1 > R$ at $\lambda = 0$, and equals 0 at $\lambda = \max_i v_i$. Hence there is a unique $\lambda^* \in (0, \max_i v_i)$ satisfying this equation. The objective is strictly convex and the feasible set is convex, so the KKT conditions are sufficient, and $\mathbf{x}^* = ((v_i - \lambda^*)_+)_{i=1}^d$ is the unique optimum. □

Algorithm:

1. If $\sum_{i=1}^d v_i \leq R$, return $\mathbf{x}^* = \mathbf{v}$.
2. Sort $\mathbf{v}$ in decreasing order: $v_{\pi(1)} \geq v_{\pi(2)} \geq \cdots \geq v_{\pi(d)}$.
3. Set $S \leftarrow 0$. For $k = 1, 2, \ldots, d$:
   (a) Set $S \leftarrow S + v_{\pi(k)}$.
   (b) Set $\lambda_k \leftarrow \frac{S-R}{k}$.
   (c) If $k = d$ or $\lambda_k \geq v_{\pi(k+1)}$, set $\lambda^* \leftarrow \lambda_k$ and break.
4. Return $x_i^* = (v_i - \lambda^*)_+$ for each i.

Correctness: We need λ^* such that $\sum_i (v_i - \lambda^*)_+ = R$. Suppose exactly the top k components survive thresholding, i.e., $v_{\pi(k)} > \lambda^* \geq v_{\pi(k+1)}$ (with $v_{\pi(d+1)} := 0$). Then

$$\sum_{i=1}^{d} (v_i - \lambda^*)_+ = \sum_{j=1}^{k} (v_{\pi(j)} - \lambda^*) = S_k - k\lambda^* = R,$$

where $S_k = \sum_{j=1}^k v_{\pi(j)}$, giving $\lambda^* = (S_k - R)/k$. The algorithm checks each candidate k in order and verifies the condition $v_{\pi(k)} > \lambda_k \geq v_{\pi(k+1)}$: since $\|\mathbf{v}\|_1 > R$, we have $\lambda_1 =$

$v_{\pi(1)} - R \le v_{\pi(1)}$, and the cumulative sum S_k grows while k grows, so λ_k eventually drops below $v_{\pi(k+1)}$ (or we reach $k = d$). At that point, $\lambda_k \ge 0$ holds because $S_k \ge R$ (the partial sum of the largest components exceeds R once enough are included). It remains to verify $\lambda_k < v_{\pi(k)}$. For $k \ge 2$: at iteration $k-1$ the algorithm did not break, so $\lambda_{k-1} < v_{\pi(k)}$, i.e., $(S_{k-1} - R)/(k-1) < v_{\pi(k)}$, i.e., $S_{k-1} - R < (k-1)\, v_{\pi(k)}$. Adding $v_{\pi(k)}$ to both sides gives $S_k - R < k\, v_{\pi(k)}$, hence $\lambda_k = (S_k - R)/k < v_{\pi(k)}$. For $k = 1$: $\lambda_1 = v_{\pi(1)} - R < v_{\pi(1)}$ since $R > 0$ (which holds because $\|\mathbf{v}\|_1 > R \ge 0$ and $\mathbf{v} \ge \mathbf{0}$ imply $R > 0$ is needed for a nontrivial problem; when $R = 0$, $\mathbf{x}^* = \mathbf{0}$ is immediate).

Runtime: Sorting takes $O(d \log d)$. The loop runs at most d iterations, each in $O(1)$. Computing the final output takes $O(d)$. Total: $O(d \log d)$.

Problem 4

(i)

Proof. ($\Rightarrow$) Suppose $\kappa^\star_{\text{diag}}(\mathbf{K}) \le \kappa$. By definition, there exists a diagonal $\mathbf{W}' \in \mathbb{S}^{d\times d}_{\succ \mathbf{0}}$ such that

$$\frac{\lambda_1\big(\mathbf{W}'^{-1/2}\mathbf{K}\,\mathbf{W}'^{-1/2}\big)}{\lambda_d(\mathbf{W}'^{-1/2}\mathbf{K}\,\mathbf{W}'^{-1/2})} \le \kappa.$$

Set $\alpha = \lambda_d\big(\mathbf{W}'^{-1/2}\mathbf{K}\,\mathbf{W}'^{-1/2}\big)$ and $\beta = \lambda_1\big(\mathbf{W}'^{-1/2}\mathbf{K}\,\mathbf{W}'^{-1/2}\big)$, so $\beta/\alpha \le \kappa$ and $\alpha > 0$. The eigenvalue bounds give

$$\alpha\mathbf{I} \preceq \mathbf{W}'^{-1/2}\mathbf{K}\,\mathbf{W}'^{-1/2} \preceq \beta\mathbf{I}.$$

Define $\mathbf{W} = \alpha\,\mathbf{W}'$, which is diagonal and positive definite. Conjugating the left inequality by $\mathbf{W}'^{1/2}$:

$$\alpha\,\mathbf{W}' \preceq \mathbf{K}, \quad \text{i.e.,} \quad \mathbf{W} \preceq \mathbf{K}.$$

For the upper bound, $\mathbf{W}'^{-1/2}\mathbf{K}\,\mathbf{W}'^{-1/2} \preceq \beta\,\mathbf{I}$ gives $\mathbf{K} \preceq \beta\,\mathbf{W}'$. Since $\beta \le \kappa\alpha$,

$$\mathbf{K} \preceq \beta\,\mathbf{W}' \preceq \kappa\alpha\,\mathbf{W}' = \kappa\,\mathbf{W}.$$

($\Leftarrow$) Suppose there exists a diagonal $\mathbf{W} \in \mathbb{S}^{d\times d}_{\succ \mathbf{0}}$ with $\mathbf{W} \preceq \mathbf{K} \preceq \kappa\,\mathbf{W}$. Conjugating by $\mathbf{W}^{-1/2}$ (diagonal, hence symmetric):

$$\mathbf{I} \preceq \mathbf{W}^{-1/2}\mathbf{K}\,\mathbf{W}^{-1/2} \preceq \kappa\,\mathbf{I}.$$

Thus $\lambda_d\big(\mathbf{W}^{-1/2}\mathbf{K}\,\mathbf{W}^{-1/2}\big) \ge 1$ and $\lambda_1\big(\mathbf{W}^{-1/2}\mathbf{K}\,\mathbf{W}^{-1/2}\big) \le \kappa$, so

$$\frac{\lambda_1\big(\mathbf{W}^{-1/2}\mathbf{K}\,\mathbf{W}^{-1/2}\big)}{\lambda_d(\mathbf{W}^{-1/2}\mathbf{K}\,\mathbf{W}^{-1/2})} \le \frac{\kappa}{1} = \kappa.$$

Since $\mathbf{W}$ is a feasible diagonal preconditioner, $\kappa^\star_{\text{diag}}(\mathbf{K}) \le \kappa$. $\square$

(ii)

Proof. Let $\kappa^\star = \kappa^\star_{\text{diag}}(\mathbf{K})$. Since $\mathbf{K} \succ \mathbf{0}$, the condition number of any conjugate $\mathbf{W}^{-1/2}\mathbf{K}\,\mathbf{W}^{-1/2}$ is at least 1, so $\kappa^\star \geq 1$.

By part (i), there exists a diagonal $\mathbf{W} \in \mathbb{S}^{d\times d}_{\succ \mathbf{0}}$ with

$$\mathbf{W} \preceq \mathbf{K} \preceq \kappa^\star\,\mathbf{W}.$$

Define $\mathbf{W}' = \mathbf{W} + \mathbf{D}$, which is diagonal and positive definite (as the sum of two diagonal positive definite matrices). We verify the two Loewner inequalities for $\mathbf{K} + \mathbf{D}$:

Lower bound: $\mathbf{W} \preceq \mathbf{K}$ implies $\mathbf{W} + \mathbf{D} \preceq \mathbf{K} + \mathbf{D}$, i.e., $\mathbf{W}' \preceq \mathbf{K} + \mathbf{D}$.

Upper bound: From $\mathbf{K} \preceq \kappa^\star\,\mathbf{W}$,

$$\mathbf{K} + \mathbf{D} \preceq \kappa^\star\,\mathbf{W} + \mathbf{D}.$$

Since $\kappa^\star \geq 1$ and $\mathbf{D} \succeq \mathbf{0}$, we have $\mathbf{D} \preceq \kappa^\star\,\mathbf{D}$, hence

$$\kappa^\star\,\mathbf{W} + \mathbf{D} \preceq \kappa^\star\,\mathbf{W} + \kappa^\star\,\mathbf{D} = \kappa^\star\,\mathbf{W}'.$$

Combining: $\mathbf{K} + \mathbf{D} \preceq \kappa^\star\,\mathbf{W}'$.

We have shown $\mathbf{W}' \preceq \mathbf{K} + \mathbf{D} \preceq \kappa^\star\,\mathbf{W}'$ with $\mathbf{W}'$ diagonal and positive definite. By part (i),

$$\kappa^\star_{\text{diag}}(\mathbf{K} + \mathbf{D}) \leq \kappa^\star = \kappa^\star_{\text{diag}}(\mathbf{K}). \qquad \square$$

Problem 5

Proof. Fix $y \in \operatorname{int}(\operatorname{dom} f^*)$ and set $x = \nabla f^*(y)$, so $y = \nabla f(x)$ with $x \in \operatorname{int}(\operatorname{dom} f)$. Write $H = \nabla^2 f(x)$. Since f is Legendre type, the conjugate satisfies

$$\nabla^2 f^*(y) = \big(\nabla^2 f(x)\big)^{-1} = H^{-1}.$$

To compute $D^3 f^*(y)[u, u, u]$ for an arbitrary $u \in \mathbb{R}^d$, introduce the path $y(t) = y + tu$ and let $x(t) = \nabla f^*(y(t))$, so that $y(t) = \nabla f(x(t))$. Differentiating at $t = 0$,

$$u = \nabla^2 f(x)\,x'(0) = H\,x'(0), \qquad \text{hence} \quad x'(0) = H^{-1}u.$$

Define $M(t) = \nabla^2 f(x(t))$, so $\nabla^2 f^*(y(t)) = M(t)^{-1}$. By the given matrix derivative formula,

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} M(t)^{-1} = -H^{-1}\left(\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} M(t)\right) H^{-1}.$$

For any $a, b \in \mathbb{R}^d$, the derivative of $M(t)$ acts as

$$a^\top\left(\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} M(t)\right) b = D^3 f(x)\big[x'(0), a, b\big] = D^3 f(x)\big[H^{-1}u,\, a,\, b\big],$$

since $M(t)_{ij} = \partial_i \partial_j f(x(t))$ and the chain rule produces the third derivative contracted with $x'(0)$.

The third derivative of f^* at y in direction u is then

$$\begin{aligned} D^3 f^*(y)[u,u,u] &= u^\top \left(\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} \nabla^2 f^*(y+tu) \right) u \\ &= u^\top \left(-H^{-1}\right) \left(\frac{\mathrm{d}}{\mathrm{d}t}\Big|_{t=0} M(t) \right) H^{-1} u \\ &= -D^3 f(x)\big[H^{-1}u,\, H^{-1}u,\, H^{-1}u\big], \end{aligned}$$

where the last equality uses the expression for $\frac{\mathrm{d}M}{\mathrm{d}t}$ with $a = H^{-1}u$ and $b = H^{-1}u$, together with multilinearity of $D^3 f(x)$.

Set $h = H^{-1}u$. Self-concordance of f gives

$$\big|D^3 f(x)[h,h,h]\big| \le 2\big(D^2 f(x)[h,h]\big)^{3/2} = 2\big(h^\top H\, h\big)^{3/2}.$$

Substituting $h = H^{-1}u$:

$$h^\top H\, h = (H^{-1}u)^\top H\, (H^{-1}u) = u^\top H^{-1} u = D^2 f^*(y)[u,u].$$

Therefore

$$\big|D^3 f^*(y)[u,u,u]\big| = \big|D^3 f(x)[h,h,h]\big| \le 2\big(u^\top H^{-1} u\big)^{3/2} = 2\big(D^2 f^*(y)[u,u]\big)^{3/2}.$$

Since $y \in \operatorname{int}(\operatorname{dom} f^*)$ and $u \in \mathbb{R}^d$ were arbitrary, f^* is self-concordant. $\square$